

7.8 Quiz: Logarithms (Version A) — Markscheme

1. [Maximum mark: 5]

$$\text{(a)} \ln 5 + \ln 6 = \ln(5 \cdot 6) = \ln 30. \quad \mathbf{A1}$$

$$\text{(b)} 2 \ln 3 = \ln 3^2 \quad \mathbf{A1}$$

$$= \ln 9. \quad \mathbf{A1}$$

$$\text{(c)} -\ln \frac{1}{4} = \ln \left(\frac{1}{4} \right)^{-1} \quad \mathbf{A1}$$

$$= \ln 4. \quad \mathbf{A1}$$

Accept also $-\ln \frac{1}{4} = -(\ln 1 - \ln 4) = \ln 4$, or $-\ln \frac{1}{4} = -\ln 4^{-1} = \ln 4$.

2. [Maximum mark: 5]

$$\log_{10} 2 = p, \log_{10} 5 = q.$$

$$\text{(a)} \log_{10} 40 = \log_{10}(2^3 \cdot 5) = \log_{10} 2^3 + \log_{10} 5. \quad \mathbf{M1}$$

$$= 3 \log_{10} 2 + \log_{10} 5. \quad \mathbf{A1}$$

$$= 3p + q. \quad \mathbf{A1}$$

$$\text{(b)} \text{ Change of base: } \log_5 4 = \frac{\log_{10} 4}{\log_{10} 5} = \frac{2 \log_{10} 2}{\log_{10} 5}. \quad \mathbf{A1}$$

$$= \frac{2p}{q}. \quad \mathbf{A1}$$

3. [Maximum mark: 5]

$$\log_{10} a = \frac{1}{2}, \quad a > 0.$$

$$\text{(a)} \quad \log_{10} \left(\frac{1}{a} \right) = -\log_{10} a. \quad \text{A1}$$

$$= -\frac{1}{2}. \quad \text{A1}$$

$$\text{(b)} \quad \text{Change of base: } \log_{100} a = \frac{\log_{10} a}{\log_{10} 100} = \frac{\log_{10} a}{2}. \quad \text{A1}$$

$$= \frac{1/2}{2} \quad \text{A1}$$

$$= \frac{1}{4}. \quad \text{A1}$$

4. [Maximum mark: 5]

$$2 \ln x = \ln 25 + 6, \quad x = pe^q, \quad p, q \in \mathbb{Z}^+.$$

$$2 \ln x - \ln 25 = 6. \quad \text{Use } m \ln x = \ln x^m: \ln x^2 - \ln 25 = 6. \quad \text{M1}$$

$$\text{Use } \ln a - \ln b = \ln \frac{a}{b}: \ln \frac{x^2}{25} = 6. \quad \text{M1}$$

$$\text{Exponentiate: } \frac{x^2}{25} = e^6 \Rightarrow x^2 = 25e^6. \quad \text{A1}$$

$$\text{Since } x > 0: x = \sqrt{25e^6} = 5e^3. \quad \text{A1}$$

$$\text{So } p = 5, \quad q = 3. \quad \text{A1}$$

Accept any valid method, including expressing $6 = 6 \ln e$ first to combine all terms as a single log before exponentiating.